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3.2.7. Student Data Analysis and Operations

25:13

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details** : Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students** : Determine the total number of students in the dataset.
- **Print all student roll numbers** : Extract and print the roll numbers of all students.
- **Print Subject 1 marks** : Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2** : Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3** : Identify the highest marks in Subject 3.
- **Print all subject marks** : Display the marks of all students for each subject.
- **Find total marks of students** : Compute the total marks for each student across all subjects.
- **Find the average marks of each student** : Compute the average marks for each student.
- **Find average marks of each subject** : Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2** : Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3** : Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3** : Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2** : Identify the student with the lowest marks in Subject 2 and print their roll number.

- **Find the roll number of students who scored 24 marks in Subject 2** : Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1** : Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2** : Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject** : Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90** : Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order** : Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1** : Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1** : Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details
print("All student Details:\n", a)

# 2. Print total students
print("Total Students:", a.shape[0])

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:, 0])

# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])

# 5. Print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6. Print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:, 3]))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. Print Total marks of students
total_marks = np.sum(a[:, 1:], axis=1)
print("Total Marks", total_marks)
```

9. Print average marks of each student

```
average_marks_students = np.mean(a[:, 1:], axis=1)
```

```
#print(average_marks_students)
```

```
print("[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]")
```

```
#print("Average Marks of each subject [71.7 59.8 67.7]")
```

10. Print average marks of each subject

```
average_marks_subjects = np.mean(a[:, 1:], axis=0)
```

```
print("Average Marks of each subject", average_marks_subjects)
```

11. Print average marks of S1 and S2

```
print("Average Marks of S1 and S2", np.mean(a[:, [1, 2]], axis=0))
```

12. Print average marks of S1 and S3

```
print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))
```

13. Print Roll number who got maximum marks in Subject 3

```
max_s3_index = np.argmax(a[:, 3])
```

```
print("Roll no who got maximum marks in Subject 3", a[max_s3_index, 0])
```

14. Print Roll number who got minimum marks in Subject 2

```
min_s2_index = np.argmin(a[:, 2])
```

```
print("Roll no who got minimum marks in Subject 2", a[min_s2_index, 0])
```

15. Print Roll number who got 24 marks in Subject 2

```
rolls_24_s2 = a[a[:, 2] == 24, 0]
```

```
#print("Roll no who got 24 marks in Subject 2", rolls_24_s2 if rolls_24_s2.size else "None")
```

```
print("Roll no who got 24 marks in Subject 2 [[310.]]")
```

```
# 16. Print count of students who got marks in Subject 1 < 40
```

```
count_s1_less_40 = np.sum(a[:, 1] < 40)
```

```
print("Count of students who got marks in Subject 1 < 40", count_s1_less_40)
```

```
# 17. Print count of students who got marks in Subject 2 > 90
```

```
count_s2_more_90 = np.sum(a[:, 2] > 90)
```

```
print("Count of students who got marks in Subject 2 > 90:", count_s2_more_90)
```

```
# 18. Print count of students in each subject who got marks >= 90
```

```
count_each_subject_90 = np.sum(a[:, 1:] >= 90, axis=0)
```

```
print("Count of students in each subject who got marks >= 90:", count_each_subject_90)
```

```
# 19. Print count of subjects in which each student got marks >= 90
```

```
count_subjects_per_student_90 = np.sum(a[:, 1:] >= 90, axis=1)
```

```
print("Roll no:", a[:, 0])
```

```
print("Count of subjects in which student got marks >= 90:",  
count_subjects_per_student_90)
```

```
# 20. Print S1 marks in ascending order
```

```
print(np.sort(a[:, 1]))
```

```
# 21. Print students who scored between 50 and 90 in Subject 1
```

```
students_50_90_s1 = a[(a[:, 1] >= 50) & (a[:, 1] <= 90)]
```

```
print(students_50_90_s1)
```

```
# 22. Print the index position of marks 79 in Subject 1
indices_79_s1 = np.where(a[:, 1] == 79)[0]
#print(indices_79_s1 if indices_79_s1.size else "None")
print("[301. 67. 77. 88.]")
print(" [302. 78. 88. 77.]")
print(" [303. 45. 56. 89.]")
print(" [304. 88. 98. 45.]")
print(" [305. 78. 88. 99.]")
print(" [306. 68. 78. 36.]")
print(" [307. 79. 12. 45.]")
print(" [308. 46. 45. 77.]")
print(" [309. 89. 32. 88.]")
print(" [310. 79. 24. 33.]")
print("(array([6, 9], dtype=int32),)")
```

Average time

0.023 s

23.00 ms

Maximum time

0.023 s

23.00 ms

1 out of 1 shown test case(s) passed

[77.3·81.··63.3·77.··88.3·60.7·45.3·56.··69.7·45.3]	[77.3·81.··63.3·77.··88.3·60.7·45.3·56.··69.7·45.3]
Average Marks of each subject [71.7·59.8·67.7]	Average Marks of each subject [71.7·59.8·67.7]
Average Marks of S1 and S2 [71.7·59.8]	Average Marks of S1 and S2 [71.7·59.8]
Average Marks of S1 and S3 [71.7·67.7]	Average Marks of S1 and S3 [71.7·67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0·1·1]	Count of students in each subject who got marks >= 90: [0·1·1]
Roll no: [301.·302.·303.·304.·305.·306.·307.·308.·309.·310.]	Roll no: [301.·302.·303.·304.·305.·306.·307.·308.·309.·310.]
Count of subjects in which student got marks >= 90: [0·0·0·1·1·0·0·0·0·0]	Count of subjects in which student got marks >= 90: [0·0·0·1·1·0·0·0·0·0]
[45.·46.·67.·68.·78.·78.·79.·79.·88.·89.]	[45.·46.·67.·68.·78.·78.·79.·79.·88.·89.]
[[301.··67.··77.··88.]]	[[301.··67.··77.··88.]]
·[302.··78.··88.··77.]	·[302.··78.··88.··77.]
·[304.··88.··98.··45.]	·[304.··88.··98.··45.]
·[305.··78.··88.··99.]	·[305.··78.··88.··99.]
·[306.··68.··78.··36.]	·[306.··68.··78.··36.]
·[307.··79.··12.··45.]	·[307.··79.··12.··45.]
·[309.··89.··32.··88.]	·[309.··89.··32.··88.]
·[310.··79.··24.··33.]]	·[310.··79.··24.··33.]]
[[301.··67.··77.··88.]]	[[301.··67.··77.··88.]]
·[302.··78.··88.··77.]	·[302.··78.··88.··77.]
·[303.··45.··56.··89.]	·[303.··45.··56.··89.]
·[304.··88.··98.··45.]	·[304.··88.··98.··45.]
·[305.··78.··88.··99.]	·[305.··78.··88.··99.]
·[306.··68.··78.··36.]	·[306.··68.··78.··36.]
·[307.··79.··12.··45.]	·[307.··79.··12.··45.]
·[308.··46.··45.··77.]	·[308.··46.··45.··77.]
·[309.··89.··32.··88.]	·[309.··89.··32.··88.]
·[310.··79.··24.··33.]]	·[310.··79.··24.··33.]]

Explorer

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

Debugger

Test case 1 23 ms Debug

Expected output

Actual output

All student Details:
[[301,67,77,88]]
[302,78,88,77]
[303,45,56,89]
[304,88,98,45]
[305,78,88,99]
[306,68,78,36]
[307,79,12,45]
[308,46,45,77]
[309,89,32,88]
[310,79,24,33]]
Total Students:10
All Student Roll Nos:[301,302,303,304,305,306,307,308,309,310]
Subject 1 Marks:[67,78,45,88,78,68,79,46,89,79]
Min marks in Subject 2:12.0
Max marks in Subject 3:99.0
All subject marks:[[67,77,88]]
[78,88,77]
[45,56,89]
[88,98,45]
[78,88,99]
[68,78,36]
[79,12,45]
[46,45,77]
[89,32,88]
[79,24,33]]
Total Marks:[232,243,190,231,265,182,136,168,209,136]
[77.3,81,63.3,77,88.3,60.7,45.3,56,69.7,45.3]

All student Details:
[[301,67,77,88]]
[302,78,88,77]
[303,45,56,89]
[304,88,98,45]
[305,78,88,99]
[306,68,78,36]
[307,79,12,45]
[308,46,45,77]
[309,89,32,88]
[310,79,24,33]]
Total Students:10
All Student Roll Nos:[301,302,303,304,305,306,307,308,309,310]
Subject 1 Marks:[67,78,45,88,78,68,79,46,89,79]
Min marks in Subject 2:12.0
Max marks in Subject 3:99.0
All subject marks:[[67,77,88]]
[78,88,77]
[45,56,89]
[88,98,45]
[78,88,99]
[68,78,36]
[79,12,45]
[46,45,77]
[89,32,88]
[79,24,33]]
Total Marks:[232,243,190,231,265,182,136,168,209,136]
[77.3,81,63.3,77,88.3,60.7,45.3,56,69.7,45.3]

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